

Computer Vision (CV)

Polytechnic University of Tirana 18-29th May, 2026

CHRISTIAN MATA, PHD
christian.mata@upc.edu



- ▶ **PART 1: Contents**
- ▶ PART 2: Schedule
- ▶ PART 3: Repository
- ▶ PART 4: Activities & Evaluation

IMAGE PREPROCESSING

X					W		
0	0	0	0	0	*		
0	2	1	2	0			
0	5	0	1	0			
0	1	7	3	0			
0	0	0	0	0			
					0.5	0.7	0.4
					0.3	0.4	0.1
					0.5	1	0.5

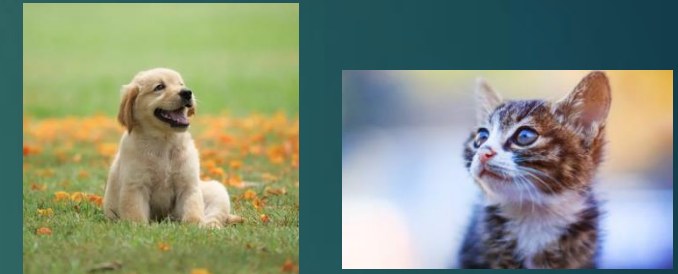
- Basic handling (math operations, reshape, logical masks)
- Filtering methods (Spatial, morphological, frequency)

IMAGE SEGMENTATION



Pixelclustering(unsupervised)
Pixel classification (supervised)
Deep Learning
(Encoder-Decoder)

IMAGE CLASSIFICATION



- Extraction of regional features
- Region-based features (Traditional ML)
- Deep Learning (CNNs)

Part 1

Module 1: Introduction to image processing (4 hours)

Basic image handling and preprocessing. Spatial (convolutional) filters, morphological operations, filters in the frequency domain.

Module 2: Image segmentation (4 hours)

Unsupervised segmentation using clustering algorithms. Supervised methods using pixelwise classifiers.

Module 3: Image classification (4 hours)

Extraction of regional features. Texture analysis and entropy. Region-based supervised classifiers. PCA Eigenfaces.

Part 2

Module 4: Introduction to deep learning (4 hours)

Convolutional neural networks for image classification. Segmentation using Convolutional Neural Networks.

- ▶ **PART 1: Contents**
- ▶ **PART 2: Schedule**
- ▶ PART 3: Repository
- ▶ PART 4: Activities & Evaluation

Computer Vision (CV)

Part 2: Schedule

TIRANA

	Monday 18	Tuesday 19	Wednesday 20		
	Presentation				
9-10h	Module 1	Module 2	Module 3		
10-11h	Module 1	Module 2	Module 3		
11-12h	Module 1	Module 2	Module 3		
12-13h	Module 1	Module 2	Module 3		
13-14h					
14-15h					
15-16h					
16-17h					
17-18h					

ONLINE

	Monday 25	Tuesday 26	Wednesday 27	Thursday 22	Friday 23
9-10h					
10-11h					
11-12h					
12-13h					
13-14h					
14-15h					
15-16h	Test 1	Module 4	Final project (no class)		
16-17h		Module 4	Final project (no class)		
17-18h		Module 4	Final project (no class)		

- ▶ **PART 1: Contents**
- ▶ **PART 2: Schedule**
- ▶ **PART 3: Repository**
- ▶ **PART 4: Activities & Evaluation**

Course materials

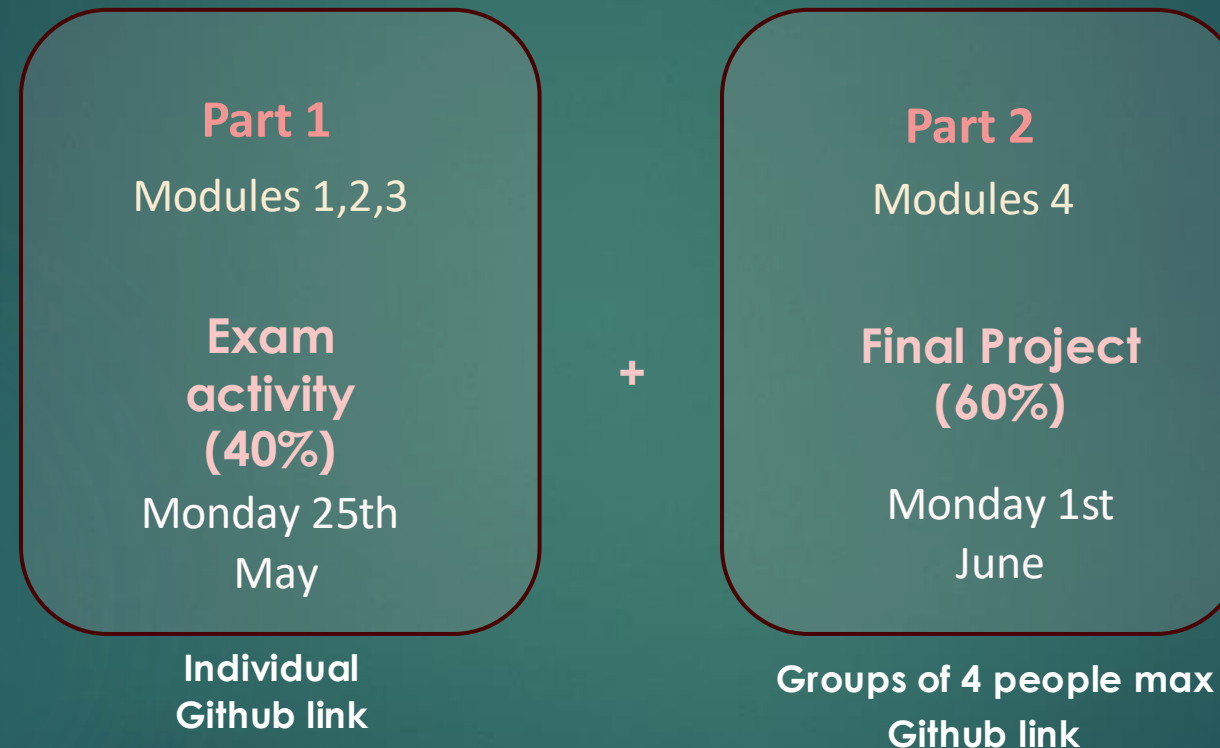


https://github.com/xtianu/CV_Tirana_2026



- ▶ **PART 1: Contents**
- ▶ **PART 2: Schedule**
- ▶ **PART 3: Repository**
- ▶ **PART 4: Activities & Evaluation**

► Activities & Evaluation



Via e-mail: christian.mata@upc.edu